Doctor Appointment System

Introduction & Purpose

This report presents the technical design of a Doctor Appointment System designed to manage doctors, their available schedules, locations, and patient appointments efficiently.
The system contains two main sides: Admin Side and Patient Side. The administrator manages doctor details, available timings, and locations. Patients can view available doctors, select a suitable location and time, and book an appointment.
The purpose of this system is to provide an organized and easy-to-use platform for managing doctor appointments and reducing manual appointment scheduling.

Scope

- Admin login
- Doctor management
- Doctor details
- Location management
- Doctor available timings
- Patient appointment booking
- Appointment management
- Database design
- Screen flow between admin and patient sides

System Overview

Admin

- Login securely
- Add doctors
- Add doctor details
- Add locations
- Add available timings
- View appointments

Patient

- View available doctors
- Select location
- Select doctor
- Select available date/time
- Enter personal details
- Book an appointment

High-Level Data Flow

Admin Login → Admin Dashboard → Add Doctor → Add Location → Add Available Timing → Patient Side → Select Location → Select Doctor → Select Timing → Enter Patient Details → Book Appointment → Appointment Stored → Admin Views Appointment

Screen Flow

- Admin Login
- ↓
- Admin Dashboard
- ↓
- Doctor Management
- ↓
- Add Doctor
- ↓
- Doctor Details
- ↓
- Add Location
- ↓
- Add Available Timing
- ↓
- View Appointments
- Patient Side
- Homepage
- ↓
- Select Location
- ↓
- Available Doctors
- ↓
- Doctor Details
- ↓
- Available Timings
- ↓
- Appointment Form
- ↓
- Appointment Confirmation

Database Design

- Admin
- Doctors
- Locations
- Doctor Timings
- Patients
- Appointments

Entity-Relationship Summary

- Admin can manage many doctors.
- Admin can manage many locations.
- One doctor can have many available timings.
- One location can have many doctors/timings.
- One patient can book many appointments.
- One doctor can receive many appointments.
- Each appointment belongs to one patient and one doctor.